AI-Enhanced Automotive Active Safety Systems

Applying ML/AI to R&D Engineering in Vehicle Dynamics & Control

NTT DATA BEST Hackathon 2026 | Politecnico di Torino

Part 1: Team & Context

Part 1

Team Introduction

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Part 1

Industry Selection & Motivation

Industry: Automotive — Active Safety & Vehicle Dynamics

Motivation: We have a working 13-DoF high-fidelity vehicle simulation with validated ESC, identifier, and ABS activation algorithms. Our research directly addresses ISO 26262 (Functional Safety) and Automotive SPICE process areas. Real impact: AI can reduce stopping distance by 21-55% on snow — this saves lives.

Part 2: Analysis & Scope

Part 2

Identified Needs & Pain Points

Blind ABS Operation

No surface knowledge, same thresholds for dry/snow/ice. 300-500ms adaptation delay at mu-jump.

Manual Calibration

10,000+ km test driving per platform. Each surface/tire combination requires separate tuning.

No Friction Data Sharing

ABS, ESC, TCS operate independently. No shared friction estimate for lateral stability.

Code Migration Burden

Algorithm validation in MATLAB, production in C/Python. Manual porting is error-prone.

Part 2

AI Tools Used

Claude (Anthropic): Algorithm design, equation derivation, activation logic, code generation, validation script, presentation

Python ML stack: scikit-learn (GP, NN, NLS), scipy (curve_fit, Brent), numpy, pandas

MATLAB/Simulink: 13-DoF vehicle simulation, 48-run validation, quadprog solver

Part 2

Process Areas (V-Model Mapping)

6: SW Design & Development (ESC algorithm, identifier, activation logic)

9: Test Definition & Planning (48-sim validation framework)

10: Test Data Management (synthetic data generation, noise models)

7: Quality Analysis (robustness sweeps, F1 metrics)

Standards: ISO 26262 (Functional Safety), Automotive SPICE (Process Assessment), ISO/SAE 21434 (Cybersecurity)

Part 3: Use Case — UC_001

Use Case ID	Short Name	Process Area	Inputs	Outputs
UC_001	AI-Enhanced Friction ID	SW Design (V-Model #6)	Wheel slip (1kHz), F_l estimates, ESC perturbation, brake torque	Burckhardt (c1,c2,c3), surface class, s_opt, mu_peak, alpha

AS IS — Current Process

Engineers calibrate ABS thresholds via 10k+ km test drives per platform. No real-time surface identification. Same parameters for all roads.

Pain Points

High cost, slow iteration, no friction data for ESC/TCS, 300-500ms blind operation at surface changes, suboptimal snow braking (+20-50% distance).

TO BE — AI-Enabled

3-layer identifier: ML-LUT (GP/NN) + ESC-gradient analytic + Brent solver. GP-ESC optimizer uses curvature for Newton-like convergence. Alpha confidence metric gates all decisions.

Expected Value

Snow stopping: -21 to -55%. Adaptation: under 100ms. s_opt convergence: 80ms (vs 4-6s). Calibration effort: -80%. Validated: 48 sims, 3 surfaces.

Commercial Tools Market Analysis

Tool	Vendor	Features	KPIs
Continental MK C2	Continental AG	Integrated ABS/ESC/TCS, hydraulic modulator, embedded ECU	ASIL-D certified, 20+ years field data
Bosch iBooster+ESP9.3	Robert Bosch GmbH	Electromech brake booster, 9th gen ESP, camera-based friction est.	Under 0.1s response, OTA updateable

Part 3: Use Case — UC_002

Use Case ID	Short Name	Process Area	Inputs	Outputs
UC_002	AI Requirements Traceability	Req. Engineering (V-Model #2)	SysML/DOORS requirements (XML), test specifications, change requests	Traceability matrix, gap report, impact analysis, compliance dashboard

AS IS — Current Process

Engineers manually link requirements to test cases in DOORS/Excel. Cross-referencing between system, SW, and HW specs done by hand. Change impact analysis takes days.

Pain Points

Missing links, inconsistent naming, manual effort 40+ hrs/release, delayed impact analysis, audit failures from incomplete traceability.

TO BE — AI-Enabled

GenAI (LLM) parses requirement documents, automatically suggests links between requirements and test cases using semantic similarity. NLP identifies gaps and inconsistencies.

Expected Value

Traceability effort: -60%. Gap detection: automated. Change impact analysis: minutes vs days. Audit readiness: continuous.

Commercial Tools Market Analysis

Tool	Vendor	Features	KPIs
IBM DOORS Next	IBM	Requirements lifecycle management, OSLC integration, impact analysis, baseline comparison	Industry standard, ASPICE compliant
Jama Connect	Jama Software	Live traceability, review center, risk management, REST API	Real-time trace, 40% faster reviews

Part 3: Use Case — UC_003

Use Case ID	Short Name	Process Area	Inputs	Outputs
UC_003	AI Code Quality & Migration	SW Development (V-Model #6)	MATLAB source (.m files), Simulink models (.slx), C target code	Migrated Python/C code, quality report, MISRA compliance check, diff analysis

AS IS — Current Process

Control algorithms developed in MATLAB/Simulink. Manual porting to C for ECU or Python for prototyping. Code review is manual, MISRA checking requires separate tools.

Pain Points

Porting errors, 2-3 weeks per migration cycle, no automated equivalence checking, MISRA violations found late, technical debt accumulation.

TO BE — AI-Enabled

LLM-assisted code translation (MATLAB to Python/C) with automated equivalence testing. AI-powered code review checks naming, structure, MISRA rules. Diff analysis between versions.

Expected Value

Migration time: -70%. Porting errors: -90%. MISRA compliance: automated. Code review effort: -50%.

Commercial Tools Market Analysis

Tool	Vendor	Features	KPIs
MathWorks Polyspace	MathWorks	Static analysis, MISRA C/C++, code proving, runtime error detection	ASIL-D qualified, zero false negatives on proven code
SonarQube + Copilot	GitHub/SonarSource	AI code suggestions, static analysis, security scanning, quality gates	80% bug detection, 55% faster reviews

Part 3: Use Case — UC_004

Use Case ID	Short Name	Process Area	Inputs	Outputs
UC_004	Synthetic Test Data Gen	Test Data Mgmt (V-Model #10)	Vehicle model params, Burckhardt coefficients, scenario definitions, noise profiles	Synthetic CSV datasets, labeled (slip,mu,surface), noise-augmented, balanced

AS IS — Current Process

Test data comes only from physical testing or single-scenario simulations. Limited coverage of edge cases. No systematic noise injection for robustness testing.

Pain Points

Expensive physical tests, incomplete scenario coverage, no noise models, bias toward common surfaces, insufficient data for ML training.

TO BE — AI-Enabled

Parametric simulation generates thousands of braking scenarios. 3-source noise model (sensor, slip, parameter drift) creates realistic conditions. Automated balancing ensures equal surface representation.

Expected Value

Test scenarios: 48 to unlimited. Cost: -95% vs physical. Coverage: all surfaces + edge cases. ML training data: 20k+ balanced rows. Noise-validated F1=0.97.

Commercial Tools Market Analysis

Tool	Vendor	Features	KPIs
AVL CRUISE M	AVL	Multi-domain vehicle simulation, co-simulation, virtual test drives	Real-time capable, HIL integration
IPG CarMaker	IPG Automotive	Virtual test driving, scenario-based testing, sensor simulation	500+ vehicle models, ADAS validation

Part 3: Process Standard Mapping

Part 3

Mapping Use Cases to Automotive Standards

Use Case	V-Model Area	ISO 26262	Automotive SPICE	Other Standards
UC_001 Friction ID	#6 SW Design	Part 3-4: SW Safety Req.	SWE.1-SWE.4	ISO/PAS 21448 (SOTIF)
UC_002 Req. Trace	#2 Requirements	Part 8: Safety Plan	SYS.2, SYS.3	ISO/IEC 29148
UC_003 Code Quality	#6 SW Dev + #7 QA	Part 6: SW Unit V&V	SWE.4, SWE.6	MISRA C:2012
UC_004 Test Data	#10 Test Data Mgmt	Part 4: Validation	SWE.5, SWE.6	IEEE 1012 (V&V)

Why Automotive Industry?

Safety-critical: ABS failure = lives at risk. AI must be validated rigorously (ASIL-D).

Data-rich: Modern vehicles generate TB/day. Untapped potential for ML-based optimization.

Standardized: ISO 26262 + A-SPICE provide clear process framework for AI integration.

Electrification: EVs need friction-aware regen braking — our framework enables this directly.

Part 4: Clustering & Prioritization Strategy

Part 4

Clustering Criteria Definition

Technical Feasibility (25%)

Can we implement this with current technology and data?

1=Research needed, 2=Prototype possible, 3=Ready to implement, 4=Already validated, 5=Production-ready

Business Impact (30%)

How much value does this create for OEMs?

1=Marginal, 2=Noticeable, 3=Significant (>10% improvement), 4=Major (>30%), 5=Transformative

Innovation Level (25%)

How novel is this vs existing market solutions?

1=Commodity, 2=Incremental, 3=New combination, 4=Novel method, 5=Paradigm shift

Strategic IP Value (20%)

Does this create defensible intellectual property?

1=No IP, 2=Know-how, 3=Trade secret, 4=Patentable, 5=Platform IP

Weighted Score = 0.25*Feasibility + 0.30*Impact + 0.25*Innovation + 0.20*IP

Part 4: Scoring Matrix & Prioritization

Part 4

Use Case Scoring Results

Use Case	Feasibility (25%)	Impact (30%)	Innovation (25%)	IP Value (20%)	Weighted Score	Recommendation
UC_001 Friction ID	4	5	5	5	4.75	MAKE
UC_004 Test Data Gen	5	4	4	4	4.25	MAKE
UC_002 Req. Trace	5	4	2	1	3.15	BUY
UC_003 Code Quality	5	3	2	1	2.85	BUY

MAKE — Why In-House?

UC_001 & UC_004 create unique IP. No commercial tool offers ESC-gradient identification or alpha-gated activation. Our 13-DoF simulation is the foundation for synthetic data generation. These are strategic differentiators — building internally preserves competitive advantage.

BUY — Why Market Tools?

UC_002 & UC_003 are commodity processes. IBM DOORS and Polyspace are industry-proven, ASPICE-compliant, and have decades of automotive validation. Building in-house would reinvent the wheel with no strategic advantage. Speed to value: immediate deployment.

Part 5: Executive Proposal

Part 5

Investment Recommendation to Top Management

Portfolio: 2 MAKE (Strategic IP) + 2 BUY (Proven Solutions) = Balanced Innovation

MAKE UC_001

AI Friction ID

Snow: -55% stopping distance

6-12 months R&D

ROI: Platform IP across all vehicles

MAKE UC_004

Synthetic Test Data

Test cost: -95%

3-6 months R&D

ROI: Platform IP across all vehicles

BUY UC_002

Req. Traceability

Trace effort: -60%

License: ~50k EUR/yr

ROI: Immediate effort reduction

BUY UC_003

Code Quality

Migration errors: -90%

License: ~30k EUR/yr

ROI: Immediate effort reduction

Part 5: MAKE — Technical Innovation

UC_001: Why No Commercial Alternative Exists

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ESC-Gradient Identifier (novel): c1 = (mu - g_true*|s|) / (1-(1+c2*|s|)*exp(-c2*|s|))
GP-ESC Optimizer (novel): ds = alpha*[-2*g_esc/(a*K_id)] + (1-alpha)*[k_esc*g_esc]
Algebraic Activation (novel): threshold = base*(1 - ln(alpha)/k_conf)   [derived from alpha error def]
Alpha confidence: alpha = LPF(exp(-k * |mu_meas - mu_est| / mu_meas))  [6 roles, 1 metric]
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Validated: 48 simulations, 8 scenarios, 3 brake distributions

21-55%	12-32%	80ms	F1=0.97
Snow stopping improvement	Dry stopping improvement	Peak convergence (vs 4-6s)	Surface classifier (with noise)

Simulation: 13-DoF vehicle, Burckhardt tire model, v0=30m/s, MATLAB/Simulink. Robustness tested with 3-source noise model.

Part 5: Business Impact & ROI

Part 5

Quantified Benefits Summary

Benefit Area	Current (AS IS)	Proposed (TO BE)	Improvement
Snow stopping distance	Baseline (lockup)	Optimized slip tracking	-21 to -55%
Surface adaptation time	300-500ms	Under 100ms	3-5x faster
ABS calibration effort	10,000+ km testing	Data-driven ML training	-80%
Req. traceability effort	40+ hrs/release	AI-assisted linking	-60%
Code migration cycle	2-3 weeks	AI-assisted porting	-70%
Test data generation	Physical testing only	Parametric + noise model	-95% cost

Strategic Value for Automotive OEMs

Retrofit-compatible: Adds ON TOP of existing ABS hardware. Classical fallback preserved (alpha=0).

EV-critical: Friction-aware regenerative braking coordination. Enables safe one-pedal driving.

Platform IP: Same framework for any vehicle. One alpha, one solver, one perturbation signal.

Use Case Definition Template

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Use Case ID	Short Name	Reference Process Area	Characteristics of Inputs	Characteristics of Outputs
UC_001	*AI-Enhanced Tire-Road Friction ID*	*Vehicle Dynamics / Active Safety (ABS)*	*Wheel slip (1kHz encoder), F_l estimates (IMU), ESC perturbation signal, brake torque*	*Burckhardt (c1,c2,c3), surface class, s_opt, mu_peak, confidence alpha*

AS IS Situation

Current Process Description

Industry ABS uses wheel deceleration thresholds: detect lockup, dump pressure, wait for re-acceleration, rebuild pressure. No surface knowledge, no optimal slip tracking, same parameters for all surfaces. Adapts to mu-jump in 300-500ms (2-3 ABS cycles). Calibration requires 10,000+ km of test driving per platform.

Current Process Criticalities and Pain Points

Blind operation (no surface knowledge), 300-500ms adaptation delay at mu-jump, suboptimal braking on snow/ice (+20-50% stopping distance), high calibration cost (10k+ km), no friction data for ESC/TCS/lateral stability, single-dimensional control (lockup yes/no only).

TO BE Situation (AI Enabled)

AI/GenAI Application Description

3-layer identifier: ML-enhanced LUT (GP/NN trained on tire data) provides coarse c2, ESC-gradient analytic solves c1/c3 at any operating point (no peak assumption), Brent refines c2 with two-point root-finding. GP-ESC optimizer uses identifier curvature for Newton-like peak convergence (~80ms all surfaces). Alpha confidence metric gates all decisions continuously.

Expected Added Value & Benefits

Snow stopping distance: -21 to -55%. Dry stopping distance: -12 to -32%. Mu-jump adaptation: less than 100ms (3-5x faster). Surface ID: F1=0.97 (noisy). Calibration effort: -80% (data-driven). s_opt convergence: 80ms vs 4-6s (50-75x faster). Validated: 48 simulations, 3 surfaces, 3 brake distributions.

Commercial Tools Market Analysis

List at least 2 potential market solutions

Tool Name	Tool Vendor	Reference Website	Tool Features	Expected Performances & KPIs
Continental MK C2	*Continental AG*	*continental-automotive.com*	*Integrated ABS/ESC/TCS, hydraulic modulator, embedded ECU, wheel speed sensors*	*ASIL-D certified, 20+ years field data, industry standard stopping distance*
Bosch iBooster + ESP 9.3	*Robert Bosch GmbH*	*bosch-mobility.com*	*Electromech brake booster, 9th gen ESP, regen braking for EVs, camera-based friction est.*	*Under 0.1s response, OTA updateable, supports brake blending*

One Signal. One Metric. One Solver.

ESC perturbation: optimizer + identifier + gradient source

Alpha: confidence metric across all 6 system roles

quadprog: single solver for QP1, QP2, and MPC

2 MAKE: Friction ID + Synthetic Test Data (unique IP, no market equivalent)

2 BUY: Requirements Traceability + Code Quality (proven tools, immediate value)

Thank you

Kaan | Sajjad Shahali | Politecnico di Torino | NTT DATA BEST Hackathon 2026

Tools used: Claude (Anthropic) — algorithm design, code generation, presentation | Python scikit-learn — ML models | MATLAB/Simulink — vehicle simulation